

Project: ManuTrack - Manufacturing Operations & Production Management Platform

1. Introduction

ManuTrack is a web-based Manufacturing Operations and Production Management Platform designed for factories, production plants, and industrial enterprises to unify production planning, work order tracking, inventory control, and compliance reporting. The platform consolidates product data, work orders, machine usage, and quality records to provide end-to-end visibility, efficient workflows, and adherence to industry standards.

It supports RESTful APIs with Angular/React frontends, and backend implementations in Java Spring Boot or ASP.NET Core. The architecture emphasizes reliability, secure handling of operational data, and compliance with manufacturing regulations.

Actors / Users

- **Production Planner:** Creates and manages production schedules and work orders.
- **Shop Floor Operator:** Executes work orders, updates progress, and reports issues.
- **Quality Inspector:** Records inspection results and manages non-conformance reports.
- **Inventory Manager:** Oversees raw material and finished goods stock.
- **Compliance Officer:** Reviews audit logs and generates regulatory reports.
- **Admin:** Configures products, machines, users, and system settings.

2. Module Overview

2.1 Identity & Access Management

Authentication, RBAC, audit trails.

2.2 Product & BOM Management

Product registration, bill of materials (BOM), and versioning.

2.3 Work Order & Production Management

Work order creation, scheduling, and execution tracking.

2.4 Inventory & Material Management

Raw material and finished goods tracking, purchase orders, and stock reconciliation.

2.5 Quality & Inspection Management

Inspection workflows, defect logging, and non-conformance handling.

2.6 Compliance & Audit Management

Audit logs, regulatory reporting, and compliance dashboards.

2.7 Analytics & Reporting

Operational dashboards, KPIs, and production reports.

2.8 Notifications & Alerts

Real-time alerts for work orders, inventory, and compliance deadlines.

3. Architecture Overview

- **Frontend:** Angular or React SPA for dashboards and workflows.
- **Backend:** RESTful services (Spring Boot / ASP.NET Core).
- **Database:** Relational DB (SQL Server/MySQL/PostgreSQL).
- **Integration:** ERP, procurement systems, and regulatory portals.
- **Security:** TLS, encryption at rest, RBAC, audit logging.

4. Module Wise Design

4.1 Identity & Access Management Module

Features

- User provisioning and role assignment
- Secure login with RBAC
- Session management and password policies
- Immutable audit logs

Entities

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)

4.2 Product & BOM Management Module

Features

- Product registration and categorization
- BOM creation and versioning
- Product lifecycle management

Entities

- **Product**(ProductID, Name, Category, Version, Status)
- **BOM**(BOMID, ProductID, ComponentID, Quantity, Version, Status)

4.3 Work Order & Production Management Module

Features

- Work order creation and scheduling
- Execution tracking and progress updates
- Work order closure workflows

Entities

- **WorkOrder**(WorkOrderID, ProductID, Quantity, StartDate, EndDate, Status)
- **WorkOrderTask**(TaskID, WorkOrderID, Description, AssignedTo, Status)

4.4 Inventory & Material Management Module

Features

- Raw material and finished goods tracking
- Purchase order creation and tracking
- Stock reconciliation

Entities

- **Inventory**(InventoryID, ProductID, LocationID, QuantityOnHand, Status)
- **PurchaseOrder**(POID, SupplierID, OrderDate, ExpectedDeliveryDate, Status)

4.5 Quality & Inspection Management Module

Features

- Inspection scheduling and execution
- Defect logging and non-conformance handling
- Quality report generation

Entities

- **Inspection**(InspectionID, WorkOrderID, Date, InspectorID, Result, Status)
- **Defect**(DefectID, InspectionID, Description, Severity, Status)

4.6 Compliance & Audit Management Module

Features

- Audit log review
- Regulatory reporting generation
- Compliance dashboards

Entities

- **ComplianceReport**(ReportID, Scope, Metrics, GeneratedDate)

4.7 Analytics & Reporting Module

Features

- Production dashboards
- KPI tracking (Yield Rate, Defect Rate, On-time Completion)

- Compliance reporting
- Trend analysis

Entities

- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)

4.8 Notifications & Alerts Module

Features

- Real-time alerts for work orders and inventory
- Escalation workflows
- Configurable notification channels (in-app, email, SMS)
- Notification history

Entities

- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

5. Deployment Strategy

- **Local / Dev:** Angular/React UI, Spring Boot/.NET backend, local RDBMS.
- **Staging:** Containerized services with synthetic product and work order data.
- **Production:** On-premise or hosted deployment with API gateway, managed RDBMS with replicas.
- **High Availability:** Multi-server deployments, automated backups, disaster recovery playbooks.

6. Database Design

Tables

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)
- **Product**(ProductID, Name, Category, Version, Status)
- **BOM**(BOMID, ProductID, ComponentID, Quantity, Version, Status)
- **WorkOrder**(WorkOrderID, ProductID, Quantity, StartDate, EndDate, Status)
- **WorkOrderTask**(TaskID, WorkOrderID, Description, AssignedTo, Status)
- **Inventory**(InventoryID, ProductID, LocationID, QuantityOnHand, Status)
- **PurchaseOrder**(POID, SupplierID, OrderDate, ExpectedDeliveryDate, Status)
- **Inspection**(InspectionID, WorkOrderID, Date, InspectorID, Result, Status)

- **Defect**(DefectID, InspectionID, Description, Severity, Status)
- **ComplianceReport**(ReportID, Scope, Metrics, GeneratedDate)
- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Production Planner Dashboard:** Work orders, schedules, progress tracking.
- **Shop Floor Operator Console:** Assigned tasks, progress updates, issue reporting.
- **Quality Inspector Dashboard:** Inspection schedules, defect logging, reports.
- **Inventory Manager Console:** Stock levels, purchase orders, reconciliation.
- **Compliance Dashboard:** Audit logs, regulatory reports.
- **Admin Console:** Product setup, BOM management, user roles, system configuration.

8. Non-Functional Requirements

- **Performance:** Handle up to **25,000 concurrent users**; work order update latency under **300 ms**.
- **Security:** RBAC, TLS, encryption at rest, GDPR compliance, immutable audit logs.
- **Scalability:** Horizontal scaling for production and inventory services.
- **Availability:** Target **99.9% uptime** for manufacturing operations.
- **Maintainability:** Modular APIs, versioning, automated migrations, CI/CD pipelines.
- **Observability:** Centralized logs, metrics, and alerting.
- **Compliance:** Configurable retention policies, audit trails, regulatory reporting formats.

9. Assumptions & Constraints

- **Phase 1 scope:** Product & BOM management, work order tracking, inventory management, quality inspections, and dashboards.
- **Excluded in Phase 1:** AI-based predictive maintenance, IoT sensor integrations, advanced mobile app features.
- **Data availability:** Product, work order, and inventory data feeds available for integration.
- **Integration:** ERP, procurement systems, and regulatory portals via APIs.
- **Implementable:** Using Java Spring Boot or ASP.NET Core, Angular/React, and SQL Server/MySQL/PostgreSQL.